Introduction to SSH (Secure Shell)

1. Overview

- **SSH (Secure Shell)** is a network protocol that allows two computers to communicate **securely** over an unsecured network.
- It ensures **confidentiality**, **integrity**, and **authentication** through **strong encryption**.
- SSH is commonly used to:
  - Log in to remote systems securely.
  - Execute commands on remote servers.
  - Transfer files between computers.
  - Install software or perform maintenance tasks on remote systems.

2. SSH Use Case and Client-Server Model

- SSH operates on a **client-server model**.
  - The **SSH Client** runs on the user's local machine.
  - The **SSH Server** runs on the remote system.
- The client initiates the connection, while the server authenticates and authorizes the user.
- SSH connections can be established via:
  - **CLI (Command Line Interface)** — commonly used.
  - **GUI (Graphical User Interface)** — tools like PuTTY or MobaXterm.

3. How SSH Works

Step 1: User Setup

- The **system administrator** creates a user account on the remote server.

Step 2: Authentication Methods

There are two main ways to authenticate in SSH:

1. Username and Password

- User connects using a combination of **username** and **password**.
- Simple to use but less secure.

2. SSH Key Pair Authentication

- Consists of two cryptographic keys:
    - **Private Key:** Stored securely on the client; never shared.
    - **Public Key:** Shared with the remote server.
- Together, they form an **SSH Key Pair**.
- Only users with a registered public key on the server can access it.

Example: Jenkins Server Setup

1. Create a **Jenkins user** on the application server.
2. Generate an **SSH key pair** on the Jenkins server.
3. Add the **public key** to the authorized_keys file on the application server.

4. SSH for Services

- Many services (e.g., Jenkins, Git, Ansible) use SSH to securely connect to other servers.
- SSH service typically runs on **Port 22**.
- **Firewall Configuration:**
    - Allow access on Port 22.
    - Restrict access to **specific IP addresses** to prevent unauthorized connections.
- SSH access should be limited, as it provides high-level system control.

5. SSH Commands and Important Files

Basic Command

ssh username@hostname

- Connects to a remote host using SSH.

SSH Directory and Files

- **~/.ssh/** — Default directory for SSH configuration and key files.
    - **id_rsa / id_ecdsa** — Private key files.
    - **id_rsa.pub / id_ecdsa.pub** — Public key files.
    - **known_hosts** — Contains server keys to verify the server identity.
    - **authorized_keys** — Contains client public keys authorized to access the server.

6. Security Best Practices

Best Practices - 1

- Prefer **SSH key-based authentication** instead of passwords.
- If passwords are used:
  - Use **strong, unique passwords** for each account.
  - Do **not reuse passwords** across systems.
  - Configure **password expiration** policies.
  - Use **password managers** for 2FA, password generation, and storage.
- **Regularly update** software or enable automatic updates.
- Avoid installing unnecessary software to minimize vulnerabilities.
- Perform **regular backups** using tools like rsync.

Best Practices - 2

- Use the **root account only for administrative tasks**.
  - Log in as a normal user, then use su or sudo for root privileges.
- **Remove or lock inactive user accounts**.
- Enforce **regular password changes**.
- **Lock accounts** after multiple failed login attempts.

📘 **Reference:** Bash & Security Cheat Sheet

Best Practices - 3

- Always configure a **firewall**.
  - Restrict network access to only required ports.
  - Close unnecessary ports to reduce attack surfaces.
- Disable SSH if it is not required.
- **Disable root login via SSH:**
  - Edit the configuration file:
  - /etc/ssh/sshd_config
  - Set the parameter:
  - PermitRootLogin no

    - Restart the SSH service after making changes.
- Always log in as a normal user and switch to root only when necessary.

7. Summary Table

Concept	Description
Protocol	Secure Shell (SSH)
Default Port	22
Purpose	Secure remote login, command execution, and file transfer
Authentication Types	Password-based, SSH Key Pair
Key Files	known_hosts, authorized_keys, id_rsa, id_rsa.pub
Default Directory	~/.ssh/
Security Recommendations	Use key-based login, disable root login, update software, use firewall

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✅ **In Summary:**
SSH provides a **secure and encrypted** way to manage remote servers and services. Following proper **authentication methods** and **security best practices** ensures protection against unauthorized access and maintains system integrity.